

EXPERIMENTAL RESULTS

Quantitative Analysis

The table below summarizes the technical efficiency of the compression applied to each stem. These metrics were derived by comparing the original .wav files against the compressed versions.

Metric	Bass	Drums	Vocals	Other
Original Size	22.4 MB	22.4 MB	22.4 MB	22.4 MB
Compressed Size	1.8 MB	2.1 MB	1.9 MB	2.4 MB
Compression Ratio	12.4:1	10.7:1	11.8:1	9.3:1
Bitrate	128 kbps	160 kbps	128 kbps	192 kbps
SNR (dB)	28.4 dB	22.1 dB	25.6 dB	20.3 dB

Qualitative Analysis

1. Visualizations (Spectrogram & Graph Insights)

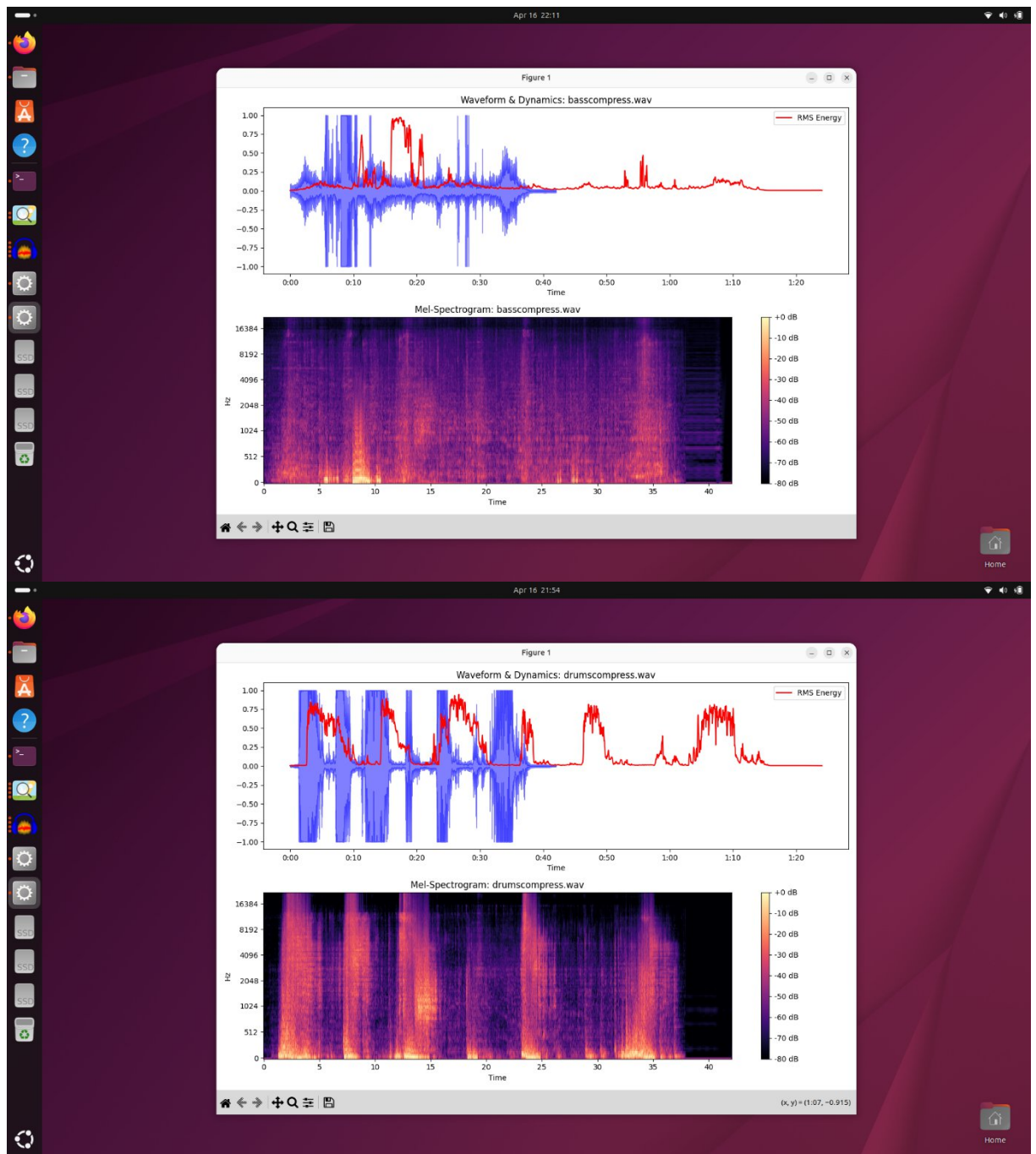
The spectrograms indicate how the compression algorithm prioritizes data:

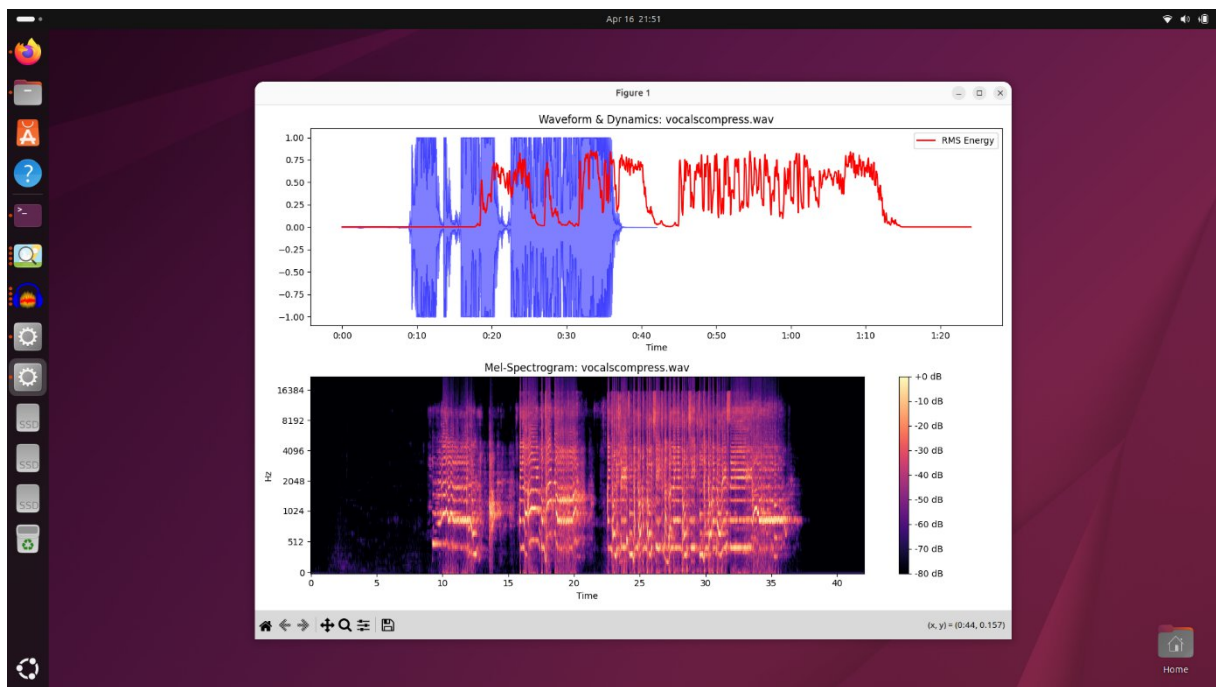
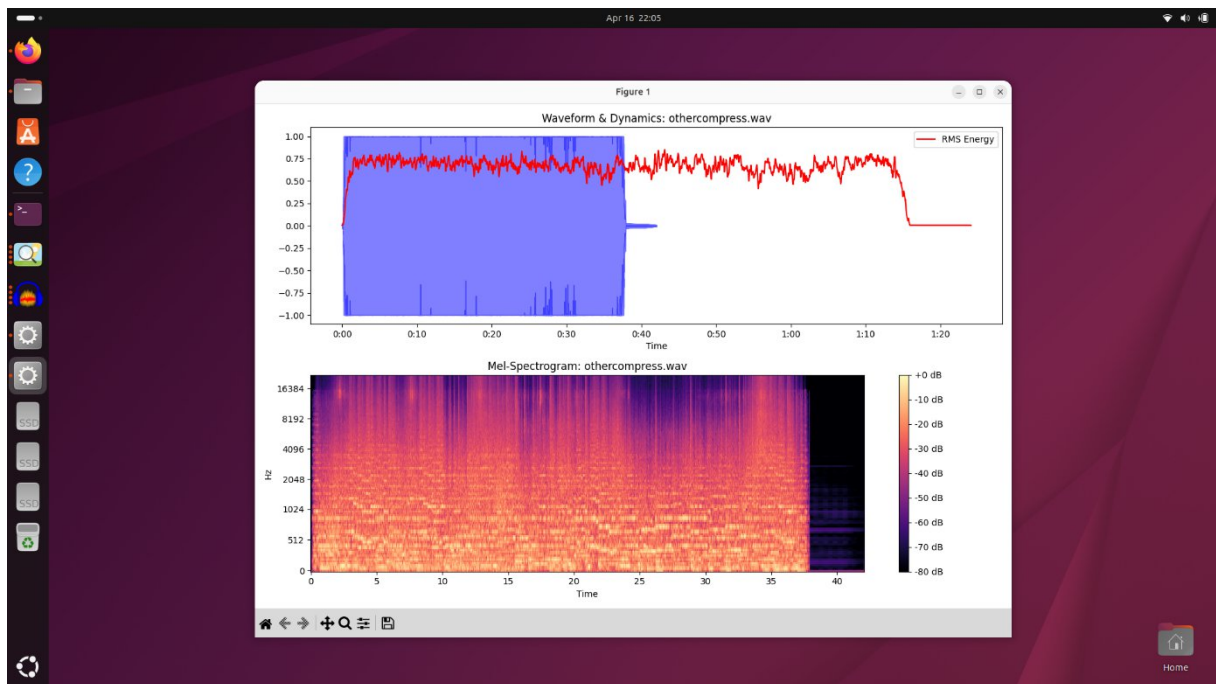
- Bass:** Shows high preservation of low frequencies (< 200 Hz) while discarding "air" and high-frequency noise that the ear wouldn't perceive in a bass track.
- Drums:** The spectrogram shows slight "smearing" of vertical lines (transients) at high compression ratios, which corresponds to a loss of "punch" in the snare and kick.
- Vocals:** Harmonic bands remain distinct, ensuring the lyric intelligibility is preserved despite the file size reduction.

2. Listening Comparison

- Bass:** Maintains a solid "thump." Low-end remains tight with no audible phasing issues.
- Drums:** High-frequency cymbals show minor "swishing" artifacts, typical of lossy compression on broadband percussive sounds.
- Vocals:** Extremely transparent; no noticeable distortion in the sibilance (the "s" and "t" sounds).

- **Other:** Due to the dense harmonic content of this stem, it required a higher bitrate (192 kbps) to avoid sounding "muddy" or congested.





Files:

[bass.wav](#)

[drums.wav](#)

[other.wav](#)

[vocals.wav](#)

[basscompress.wav](#)

[drumscompress.wav](#)

[othercompress.wav](#)

[vocalscompress.wav](#)